Mechanistic Estimation of Neural Networks: Deep Third-Cumulant Representation

Q1: Representation of the Deep Third-Cumulant Content

The fundamental wall in V29 is that the Khatri-Rao product of two low-rank matrices inflates the rank: while A_s and P_s have $r \sim 150$ effective directions, $A_s \circ P_s$ has $r^2 \sim 22,500$ directions, requiring $O(n^2 r)$ FLOPs to materialize per source. However, because the legs of an old source are transported solely by left-multiplication—$P_s \leftarrow W\mathsf{diag}(w_1)P_s$—their internal coordinates in the comoving basis *never change*.

The solution is the **Latent-Space Pooled Cumulant Tensor**, which synthesizes hints (a) and (d). It collapses the sum over all old sources into a single static tensor in the latent space, absorbing the ReLU non-linearity entirely in the $O(n)$ basis transport.

1. The Mathematical Object Carried

Instead of maintaining S separate old sources, carry exactly two objects for the entire old tier:

- A single shared transport basis $Q^{(l)} \in \mathbb{R}^{n \times r}$.
- A single *static* pooled Tucker core $T^{old} \in \mathbb{R}^{r \times r \times r}$.

2. The Per-Layer Update Rule

When a young source reaches age 5 at layer l, its dense legs are projected onto Q: $F_{A,s} = Q^T A_s$, $F_{P,s} = Q^T P_s$. We form its static core $T_s(a,b,c) = \sum_j F_{A,s}(a,j) F_{P,s}(b,j) F_{A,s}(c,j)$ and permanently add it to the pool:

$$T^{old} \leftarrow T^{old} + T_s$$

During the forward pass, T^{old} is completely static. We only update the basis to absorb the Hermite ReLU transport:

$$Q^{(l+1)} = W^{(l+1)}\mathsf{diag}(w_1^{(l)})Q^{(l)}$$

To evaluate the contribution to the slices D_{21}^{old} and D_3^{old} for the next ReLU closure, we compute the contraction of the pooled tensor directly against the basis and the birth-side nonlinearity w_2:

$$M_{ic} = \sum_{a,b} Q_{ia} Q_{ib} T^{old}_{abc}$$

$$D_{21,old} = \mathsf{diag}(w_2) M Q^T$$

(Note: If Q requires re-orthonormalization $Q \to QR$ to maintain condition number, the core updates as $T^{old} \leftarrow T^{old} \times_{1,2,3} R^{-1}$.)

3. Cost in Units Per Layer

- **Basis transport** (n^2r): 0.15 units at $r = 150$.
- **Core contraction** (nr^3 to form M, n^2r to form D_{21}): 3.2 units at $r = 150$ (10.7 units at $r = 224$).
- **Total marginal cost:** ~ 3.4 units per layer for the *entire* old tier (down from V29's 107 units).

4. What it Loses vs. Exact Transport

Algebraically, this loses **zero** precision relative to computing the per-source Tucker expansions. Summing the cores before contraction is a mathematically exact identity: $\sum_s (Q \otimes Q) T_s Q^T \equiv (Q \otimes Q)(\sum_s T_s) Q^T$. The only loss relative to exact dense transport is the truncation to rank r, which V29 already accepts.

5. First Falsifying Experiment

In a local CPU run, extract the static $F_{A,s}, F_{P,s}$ factors of the 10 old sources at layer 14 from V29. Compute their individual D_{21} contributions via the standard dense route $(A_s \circ P_s \circ w_2) A_s^T$ and sum them. In parallel, form $T^{old} = \sum_s T_s$ and evaluate $\text{diag}(w_2)(Q \circ Q) T^{old} Q^T$. The two $n \times n$ matrices must match to float64 machine precision.

Q2: Hypotheses for the Leaders' Method

Given that the leaders' bill (133–165 units) exactly equals V29's cost without the old tier (~ 154 units), they must be computing the deep third-cumulant content for strictly ≤ 10 marginal units.

Hypothesis 1: The Pooled Tensor (Rank-Tuned)

The leaders are running exactly the Latent-Space Pooled Tensor derived in Q1. By realizing that old sources can be bundled offline into a single $O(r^3)$ object, they reduced the old tier cost from $O(S \cdot n^2 r)$ to $O(1 \cdot nr^3)$. If they tune the shared rank to $r \approx 128 - 150$, the pooled tier costs roughly 2 to 4 units per layer. Added to the 154-unit young-tier base, this yields 156–158 units, dropping perfectly into the 133–165 unit bounds.

- **Observable Signature:** Their runtime will scale exactly as $O(nr^3)$ rather than linearly with depth or number of sources. On the leaderboard, they will exhibit a strictly flat per-layer cost after layer 5.

Hypothesis 2: Parity-Aware Quenched Covariance Regeneration

They developed a memoryless layer-local regeneration law for $D_{21,old}$ that respects the strict parity constraint ($E[H_s|C] = 0$). Since the deep off-diagonals must be an odd function of $W^{(l)}$, they project the covariance through the weight matrix to form a reaction field. For instance, they use a learned scalar recurrence:

$$D_{21,old}^{(l+1)} \approx \alpha_l W^{(l+1)} \text{diag}(w_1) D_{21,old}^{(l)} + \beta_l (W^{(l+1)} C^{(l)})$$

- **Observable Signature:** The marginal cost of this operation is exactly 1 unit (one $n \times n \times n$ matmul for $W \cdot C$). Falsify locally by regressing the exact V29 $D_{21,old}$ at layer 14 against $W^{(14)}C^{(13)}$ and the transported D_{21} slice; if $R^2 > 0.95$, this is their method.

Hypothesis 3: Young-Tier Shadowing (Asymptotic Alignment)

Deep in the network, the repeated application of random Jacobians causes the principal vectors of the old sources to align with the oldest of the *young* sources (age 4). Instead of carrying old sources explicitly, the leaders simply inflate the amplitude of the age-3 and age-4 young sources using a scaling scalar λ_l to "shadow" the deep tail of the cumulant.

- **Observable Signature:** Zero rank-dependent cost; old tier cost is strictly 0 units. Falsify locally by measuring the cosine similarity of the principal subspace of $D_{21,old}$ against $D_{21,\text{age}=4}$. If they are highly aligned (cosine > 0.9), the old sources have lost their independent directional information and act only as a magnitude multiplier on the young sources.